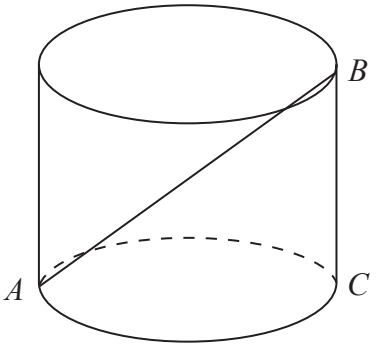
12 The diagram shows a metal rod, AB, resting inside a cylindrical tin.



The tin is on a horizontal table.
AC is a diameter of the base of the tin.
B is on the top edge of the tin.
BC is vertical.

The radius of the base of the tin is 5 cm.
The volume of the tin is 1178 cm^3

Find the angle between the rod and the base of the tin.
Give your answer correct to the nearest degree.

Cylinder volume: $\pi \times 5^2 \times h = 1178$.

1 mark
Cylinder volume

$h = 1178 \div (25\pi) = 14.99876184...$ cm.

1 mark
Find vertical height

AC = 2 × radius = 10cm; BC=h.

tan(angle BAC) = h ÷ 10.

1 mark
Opposite ÷ adjacent

Angle BAC = $\tan^{-1}(14.99876184... \div 10) \approx 56°$.

1 mark
56° nearest degree

°

(Total for Question 12 is 4 marks)